

Interface Capability Brief

EDI 276/277 Claim Status · Stakeholder overview · 2026-08-13

1. Executive summary

Ingest X12 276 claim status requests, look up a demo catalog by trace number (no SQL), stage a ClaimStatusDecision, then emit an X12 277 response and file buckets (found / not-found / error). Demo only — not a clearinghouse.

2. Who this is for

- Source: provider drop folder input/inbound/.edi - Destinations: output/found, output/not-found, output/error, output/responses/.edi (277), debug XML - Catalog is in-XSLT by TraceNumber — no claim master, no FHIR

- Source: provider drop folder input/inbound/*.edi
- Destinations: output/found, output/not-found, output/error, output/responses/*.edi (277), debug XML
- Catalog is in-XSLT by TraceNumber — no claim master, no FHIR

3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: DirectoryListener
- Format: X12 276 (005010X212), TableData 276-A1
- Identity: TRN02 / BHT03 → TraceNumber; NM1ILNM109 → MemberId; REF*EJ → ClaimId
- Samples: samples/X212-276-claim-request.edi (ABCXYZ1/2/3 in one ST — fork is //Transaction, first TRN wins)

Integration highlights

Catalog: ABCXYZ1 found/F1; ABCXYZ2 found/P1; ABCXYZ3 not-found/E1; incomplete identity → error/MISSING_IDENTITY.

Stage	Module	Notes
R1 Listener	DirectoryListener	Poll inbound; Move → archive
R1 Format	TableData 276-A1 + fork //Transaction	Extract is a target processor
R1	Pretty-print → debug txn XML → XSLT catalog → pretty-print → debug decision	Unique Pretty-Print names
R1 Transport	DirectoryTransport	Stage output/staged-decisions/
R2 Listener	DirectoryListener	Poll staged XML; Delete
R2 Router	XPathRoutingModule accumulate=true	277 when DecisionBucket present; buckets by value
R2 277	XSLT → XCSDData → pretty debug XML → XML to EDI 277-A1	Do not hardcode ISA* as text
R2 buckets	Pretty-print then DirectoryTransport	Unique names per bucket

4. Honest boundaries (not claimed yet)

- Production hardening (HA, disaster recovery, full audit workflows) is outside this demo brief
- Anything not listed in §3 should be treated as future work unless confirmed in DESIGN.md

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Intake And Lookup” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

6–9. State / validation / risks

File theater only. Unparseable EDI fails closed. First TRN per ST only. Synthetic 277 is not clearinghouse-certified.

6–9. State / validation / risks

File theater only. Unparseable EDI fails closed. First TRN per ST only. Synthetic 277 is not clearinghouse-certified.

3. Inbound contract

- Transport: DirectoryListener - Format: X12 276 (005010X212), TableData 276-A1 - Identity: TRN02 / BHT03 → TraceNumber; NM1ILNM109 → MemberId; REF*EJ → ClaimId - Samples: samples/X212-276-claim-request.edi (ABCXYZ1/2/3 in one ST — fork is //Transaction, first TRN wins)

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Stage Claim Status Decision.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Stage Claim Status Decision.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Conditional Router, Emit 277 Response, Write Found.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Emit 277 Response.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

1 - Intake And Lookup: runs steps such as Poll EDI 276 Files, Fork - XPath, Extract 276 Claim Status Fields, Pretty-Print Forked 276 XML.

2 - Emit 277 And Bucket: runs steps such as Poll Staged Claim Status Decisions, Extract Decision Attributes, Conditional Router, Map Decision To 277 EDI XML.

7. Security & trust posture

- Credentials and passwords in this package are demo-only — not production secrets

8. Seeing it work

- - Ports: EIP 8126, Web UI 8127 (no SQL) - LAN: `http://192.168.68.62:8127/` - TableData: `../../../../EDI/TableData/x12` → `edi-tabledata` - Heap: 512M–2G
- `docker compose --profile full up -d --build`
- | Web UI | `http://127.0.0.1:8127/` |
- | EIP | `http://127.0.0.1:8126/eip/` |

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.